Solution

(i) $\sqrt[3]{108} = \sqrt[3]{27 \times 4}$

$= \sqrt[3]{3^3 \times 4}$

$= \sqrt[3]{3^3} \times \sqrt[3]{4}$ (Laws of radicals - ii)

$= 3 \times \sqrt[3]{4}$ (Laws of radicals- i)

2	108
2	54
3	27
3	9
3	3
	1

order = 3; radicand = 4; coefficient = 3

(ii) $\sqrt[3]{(1024)^{-2}} = \left[\sqrt[3]{\left(2^3 \times 2^3 \times 2^3 \times 2\right)^{-2}}\right]$

$= \left[\sqrt[3]{\left(2^3 \times 2^3 \times 2^3 \times 2\right)}\right]^{-2}$ [Laws of radicals - (i)]

$= \left[\sqrt[3]{2^3} \times \sqrt[3]{2^3} \times \sqrt[3]{2^3} \times \sqrt[3]{2}\right]^{-2}$ [Laws of radicals – (ii)]

$= \left[2 \times 2 \times 2 \times \sqrt[3]{2}\right]^{-2}$ [Laws of radicals – (i)]

$= \left[8 \times \sqrt[3]{2}\right]^{-2} = \left[\frac{1}{8}\right]^2 \times \left(\frac{1}{\sqrt[3]{2}}\right)^2$

$= \frac{1}{64}\sqrt[3]{\frac{1}{4}}$

2	1024
2	512
2	256
2	128
2	64
2	32
2	16
2	8
2	4
2	2
	1

order $= 3$; radicand $= \frac{1}{4}$; coefficient $= \frac{1}{64}$

(These results can also be obtained using index notation).

Note

Consider the numbers 5 and 6. As $5 = \sqrt{25}$ and $6 = \sqrt{36}$

Therefore, $\sqrt{26}$, $\sqrt{27}$, $\sqrt{28}$, $\sqrt{29}$, $\sqrt{30}$, $\sqrt{31}$, $\sqrt{32}$, $\sqrt{33}$, $\sqrt{34}$, and $\sqrt{35}$ are surds between 5 and 6.

Consider $3\sqrt{2} = \sqrt{3^2 \times 2} = \sqrt{18}$, $2\sqrt{3} = \sqrt{2^2 \times 3} = \sqrt{12}$

Therefore, $\sqrt{17}$, $\sqrt{15}$, $\sqrt{14}$, $\sqrt{13}$ are surds between $2\sqrt{3}$ and $3\sqrt{2}$.

2.6.3 Four Basic Operations on Surds

(i) Addition and subtraction of surds : Like surds can be added and subtracted using the following rules:

$a\sqrt[n]{b} \pm c\sqrt[n]{b} = (a \pm c)\sqrt[n]{b}$, where $b > 0$.

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